

Large-Scale Analysis of New York City High Volume For-Hire Vehicle Trip Records

MIT 805 Big Data — Part 1: Data Collection and Analysis

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Repository: <https://github.com/caie007/savannahgishon> | <https://github.com/Sav-lets/>

1 Dataset provenance, licence and suitability

We analyse the *High Volume For-Hire Vehicle* (HVFHV) trip records published by the New York City Taxi and Limousine Commission (TLC), covering every trip dispatched by operators licensed under New York’s high-volume for-hire rules and issued as monthly Apache Parquet files since February 2019.¹ It is released under the NYC.gov Terms of Use as part of NYC Open Data, permitting free reuse including academic research with attribution to the City of New York, and is therefore suitable for this project without further permission. It carries no rider, driver or vehicle identifiers and generalises location to administrative zones (GPS discontinued July 2016), so no personally identifiable information is redistributed. The TLC states it did not create the data and makes no representation as to accuracy, records being base-submitted; we treat this as an analytical constraint tested in §3, not a barrier to use.

2 Scale and characteristics

The three required scales, measured on 3 September 2026, are: **raw** 37.11 GB (88 monthly files, 2019-02 to 2026-05); **working** 13.32 GB (29 files, 2024-01 onward); and **processing** 5.55 GB (12 files, calendar 2025), equal to 54.45 GB uncompressed. Table 1 gives the full breakdown. Rather than downloading the archive to size it, we issued an HTTP HEAD request against each of the 88 published files and summed **Content-Length** — an exact byte count for the full dataset, obtained without downloading it.

The processing tier holds 243,589,684 records across 25 columns. Each row is one completed trip carrying four timestamps (request, driver on-scene, pickup, dropoff), origin and destination zone identifiers, distance, duration, seven itemised monetary fields including driver pay, and five categorical service flags. We selected 2025 as the first complete year under New York’s central business district congestion pricing and the first carrying `cbd_congestion_fee`. Parquet compresses the tier 9.81×, so 5.55 GB on disk becomes 54.45 GB in memory; the latter governs processing feasibility.

3 Preparation and data quality

Schema. Verified directly rather than from documentation: June 2019 carries 24 columns and June 2025 carries 25, the sole addition being `cbd_congestion_fee`. The series is substantially schema-stable and `mergeSchema` handles the 2025 boundary; we restricted the working tier to 2024 onward so every record within it is structurally identical.

Missing data. Two of 25 fields carry nulls (Fig. A1). `originating_base_num` is absent for 67,385,046 records (27.66 %) — structurally inapplicable where no distinct originating base exists, rather than erroneously omitted. `on_scene_datetime` is null for 10,870,353 (4.46 %); because completeness varies by operator rather than at random, driver-arrival analysis is implicitly weighted toward operators that report it. All monetary and service-flag columns are complete.

Validity. We audited internal consistency rather than trimming extremes, since a long trip is not necessarily erroneous whereas a dropoff preceding its pickup certainly is. Nine of ten checks return under 0.025 % (Table A1); only 94 records in 243,589,684 imply speeds above 90 mph. One check dominates: 2,790,424 records (1.15 %) show pickup preceding request, a logically impossible ordering we attribute to clock drift between operator telematics and the TLC pipeline. These are excluded from wait-time analysis only and retained elsewhere, rather than discarding 2.8 million valid fare records for a timestamp fault.

Duplicates. With no trip identifier, duplication was assessed on a composite key of licensee, both timestamps, both zones, fare and driver pay: 26 repeats in 243,589,684 (1×10^{-5} %). In a city dispatching over 600,000 trips daily these are plausibly coincident genuine trips; we retain them, and their near-absence bounds the TLC’s accuracy disclaimer.

Scope restriction. We retain only trips with both endpoints among the 265 mapped zones, excluding 11,126,292 records (4.57 %) and keeping 232,463,392. This is a scoping decision, not a quality filter, and it is not neutral: excluded trips average 16.74 miles and \$70.08 against 4.48 miles and \$24.90 retained, and are 99.9 % dropoff-side — a long-haul regional segment, not corruption. Exclusion makes zone-level analysis tractable but biases distance and fare downward.

4 Exploratory data analysis

Market structure. The schema accommodates four licensees but only two operated in 2025; Juno and Via record no trips. Uber dispatched 167,756,559 trips (72.16 %) against Lyft’s 64,706,833 (27.84 %), a Herfindahl–Hirschman index of 5,982. Yet the two are near-indistinguishable on what a passenger observes: fares differ 8 % (\$25.47 versus \$23.41) and median wait by 16 seconds (237s versus 253s), with Lyft better at p90 (495s versus 505s). A 2.59:1

¹ <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>; mirrored at AWS Open Data Registry `s3://nyc-tlc`. A companion CSV maps the 265 taxi zone identifiers to zone and borough names.

volume advantage resting on a service edge of seconds implies the duopoly is sustained by demand-side factors the data cannot observe, not by observable differentiation.

Temporal structure. Volume averages 636,886 trips per day, ranging 495,548 to 887,777 (Fig. A2). The extremes are structurally interpretable: the busiest days are December weekends plus Halloween Saturday, the quietest are public holidays — Christmas (501,610), Labor Day (501,250), Memorial Day (498,048) — so December holds both the annual maximum and near-minimum within twelve days. Across the week demand varies 9.3-fold (Fig. A3), from 2,283,178 trips at Saturday 19:00 to 245,562 at Tuesday 03:00. Less obviously, the morning commute reaches 91 % of the weekend peak and exceeds the evening one: Wednesday 08:00 records 2,079,169 against 18:00's 2,015,243.

Service quality. Median wait is 4.0 min (p90 8.3 min), but its shape inverts intuition (Fig. A4): wait peaks at 5.9 min at 04:00 and troughs at 4.05 min at 10:00, so passengers wait longest when demand is lowest — the binding constraint is driver supply, not congestion. The paired flags expose unmet demand rarely visible in open data: of 6,889,088 shared-ride requests only 3,939,396 matched (57.18 %), so 2,949,692 passengers accepted a discounted shared fare and rode alone.

Geography and economics. The busiest zones are airports — LaGuardia (4,250,005 trips, \$51.71 mean fare) and JFK (3,247,011, \$62.52) — whose signature recurs as a secondary mode near 17 miles in the distance distribution (Fig. A5). Below them outer-borough residential zones dominate: Crown Heights North (3,138,991) outranks every Manhattan zone and six of the top fifteen are in Brooklyn (Fig. A6). Fare inverts this ordering — East New York \$18.01 and Crown Heights \$20.77 against Midtown \$36.35 and Times Square \$38.43 — indicating two segments: high-frequency, low-value trips where rail coverage is weak, and higher-value discretionary travel in the core. Trips stay within one borough 72.21 % of the time, the largest cross-borough flow being just 4.01 %. Driver pay averages 77.24 % of base fare and is stable rather than trending, holding within 76–79 % for ten of twelve months with an unexplained March trough (73.28 %) and October peak (79.48 %); December closes at 77.58 %. Tips are flat at 3.9 % to 4.8 % (Fig. A7).

5 The seven Vs of big data

Volume. 37.11 GB compressed, $\approx$ 364 GB uncompressed; 243,589,684 records in the processing year. Volume binds rather than describes: the duplicate check required a full shuffle over 244 million rows, taking seven minutes across eight cores. **Velocity.** 636,886 trips per day, 7.4 records generated per second sustained across the year — but published monthly with a two-month lag. This gap between a real-time generating process and a batch-published archive is the dataset's defining constraint: it supports retrospective policy analysis but forecloses the real-time dispatch applications its timestamp structure superficially suggests. **Variety.** 25 columns spanning four timestamp fields, categorical licensee and base codes, integer zone keys requiring joins to an external CSV and shapefile, decimal currency fields and Y/N flags — the licensee field itself a four-way categorical that resolves to two in practice. **Value.** See §6. **Veracity.** Our audit both substantiates and bounds the TLC's disclaimer. The 1.15 % of records with pickup preceding request is a real systematic fault; against that, duplication is effectively absent and nine of ten checks return below 0.025 %. A further 531 records carry negative driver pay (minimum $-\$55.02$) against a high mean fare of \$76.62, consistent with retrospective adjustments netted into the record. The monetary fields are therefore a settled accounting position, not the amount transacted at the time of travel. **Variability.** Figure A8 shows the 2020 collapse, monthly volume falling from roughly 500 MB to 110 MB, and a recovery now nearly complete: June 2025 recorded 19,868,009 trips against 21,001,990 in June 2019, or 94.6 % of pre-pandemic volume. That recovery is not discernible from file size alone, since bytes per row are near-identical across the two periods (24.7 versus 25.5) — a caution against inferring record counts from storage footprint. **Volatility.** Records persist from 2009, so the data is durable in storage, but its decision relevance decays far faster: the publication lag and the January 2025 regime change leave earlier behavioural patterns with limited operational shelf life.

6 Value and limitations

Value. The ratio of `driver_pay` to `base_passenger_fare` makes the effective driver share directly observable — the evidentiary basis for TLC minimum-pay rulemaking. Across \$5.79 billion in 2025 base fares drivers received \$4.47 billion (77.24 %), so one percentage point of share is worth roughly \$57.9 million annually — the dataset quantifies the exact lever those rules operate on, without access to operator accounts. For transport policy, `cbd_congestion_fee` shows the levy applied to 78,083,012 trips (33.59 %), raising \$117.1 million at a mean of \$1.500002 — a flat per-trip charge, and therefore regressive: 7.2 % of the average Crown Heights fare against 2.4 % at JFK, falling hardest on the outer-borough segment identified above as transit-substituting. For accessibility, 685,364 wheelchair-accessible requests met 685,353 matches; we read a near-perfect rate on a capacity-constrained fleet as reflecting the TLC's dispatch mandate rather than an unconstrained market. The revealing figure is the request rate itself — 0.295 % of trips, below any plausible estimate of the wheelchair-using travelling population.

Limitations. *Survivorship.* Only matched, completed trips appear; cancellations, unfilled requests and the surge multiplier are absent. The data measures served demand, not demand: a zone with few trips may be underserved rather than low-demand, and the dataset cannot distinguish the two — precisely the distinction a regulator needs. The same limit governs the accessibility figures: requests made are recorded, journeys forgone are not. *No entity identifiers.* Without driver, vehicle or rider keys, no analysis of earnings per shift, utilisation or retention is possible; all inference is trip-level. This is an appropriate privacy protection but forecloses the labour-economics questions the pay fields otherwise invite. *Jurisdictional coverage.* Manhattan-to-Newark trips are 0.61 % of the total while Newark-origin trips are absent, return journeys falling under New Jersey licensing. The boundary is regulatory, not spatial.

Appendix

Table 1: Dataset tiers, measured 3 September 2026. Uncompressed size for the processing tier is computed from its schema and row count; raw and working are scaled by the resulting $9.81\times$ ratio.

Tier	Period	Files	Parquet (GB)	Uncompressed (GB)
Raw	2019-02 – 2026-05	88	37.11	≈ 364
Working	2024-01 – 2026-05	29	13.32	≈ 131
Processing	2025-01 – 2025-12	12	5.55	54.45

The processing

Table 2: Validity audit, 2025 processing tier (243,589,684 records).

Check	Records	% of tier
pickup before request	2,790,424	1.1455
driver_pay ≤ 0	58,781	0.0241
trip_miles ≤ 0	28,128	0.0115
trip_miles > 100	22,397	0.0092
base_passenger_fare < 0	13,495	0.0055
dropoff $\leq$ pickup	12,231	0.0050
unknown pickup zone (264/265)	10,398	0.0043
trip_time > 6 h	465	0.0002
implied speed > 90 mph	94	0.0000
trip_time ≤ 0	31	0.0000

Table 3: Summary statistics, retained records (232,463,392).

Field	Mean	Std. dev.
trip_miles	4.48	4.43
trip_time (s)	1141.1	787.1
base_passenger_fare	24.90	19.64
driver_pay	19.26	14.67
tips	1.07	3.07
tolls	0.73	2.57
congestion_surcharge	1.02	1.32
airport_fee	0.21	0.70

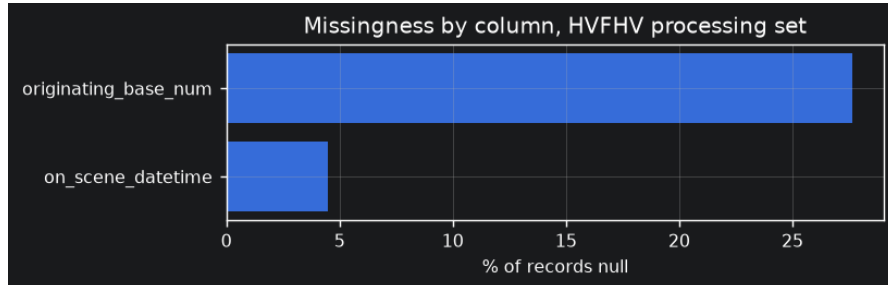


Figure A1. Missingness by column; only two of 25 fields contain nulls.

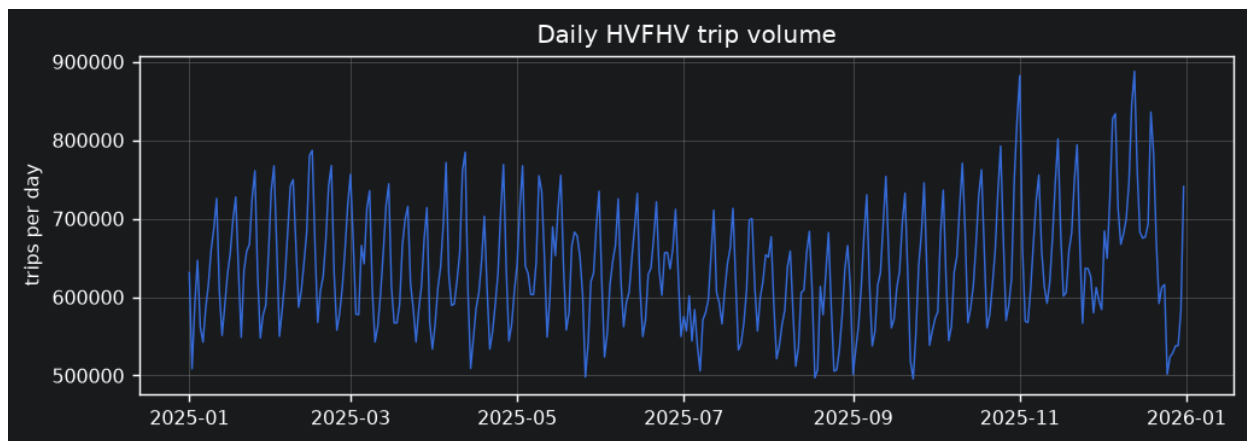


Figure A2. Daily trip volume, 2025.

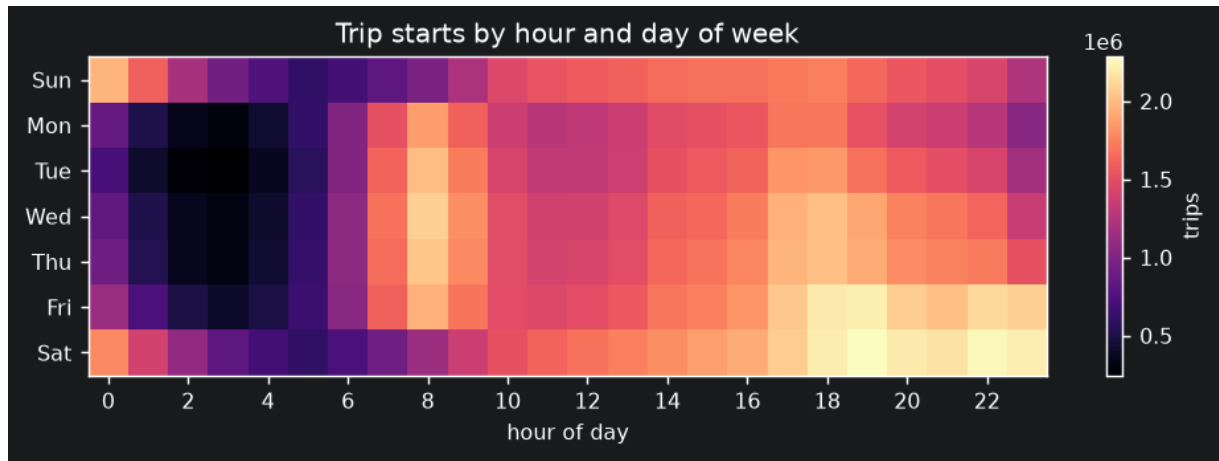


Figure A3. Trip starts by hour of day and day of week. The Saturday evening peak, the weekday morning band and the Tuesday pre-dawn trough are all visible.

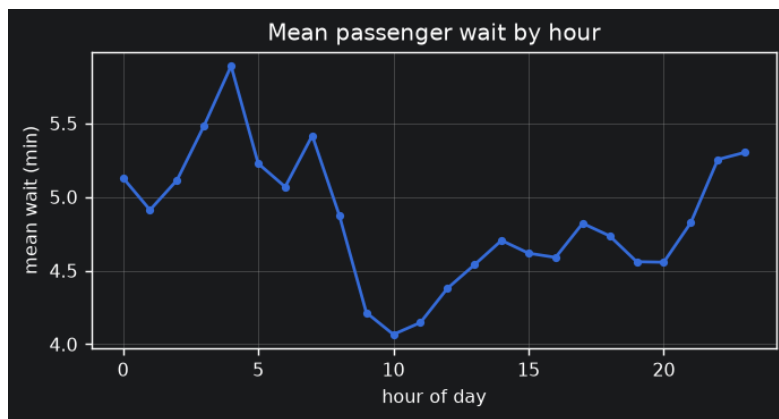


Figure A4. Mean passenger wait by hour — longest at 04:00, shortest at 10:00, the inverse of the demand profile.

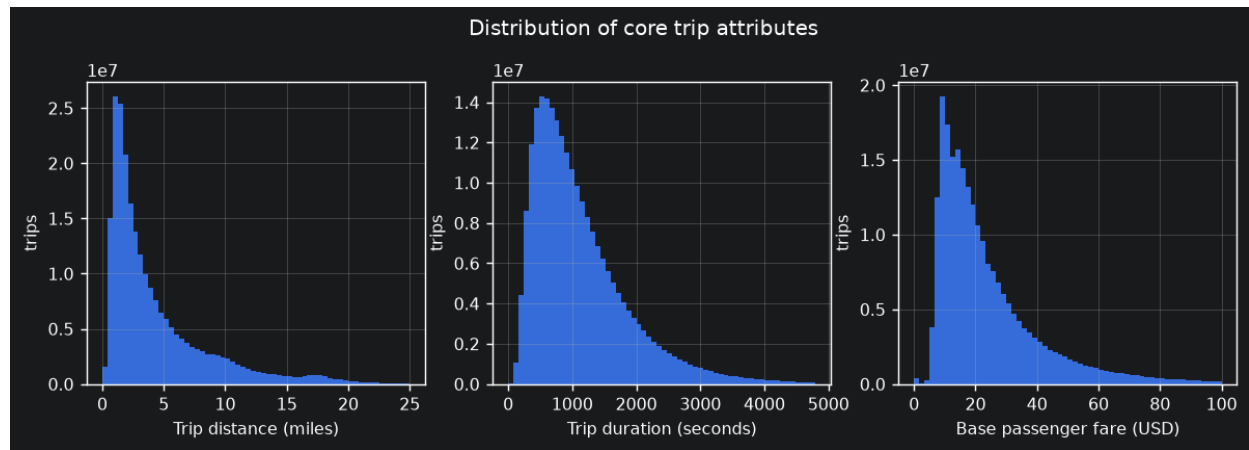


Figure A5. Distance, duration and fare distributions, truncated at 25 miles, 4800 seconds and \$100. Note the secondary mode near 17 miles corresponding to airport trips.

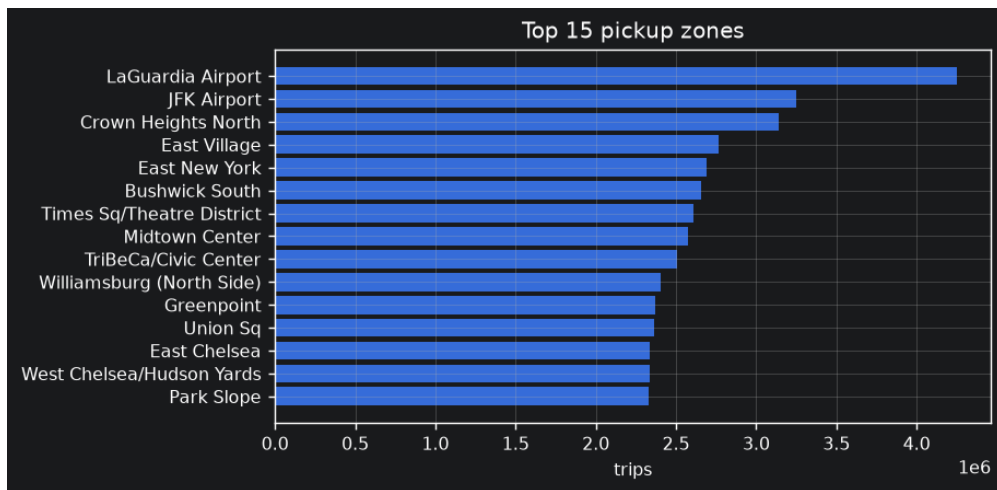


Figure A6. Fifteen busiest pickup zones.

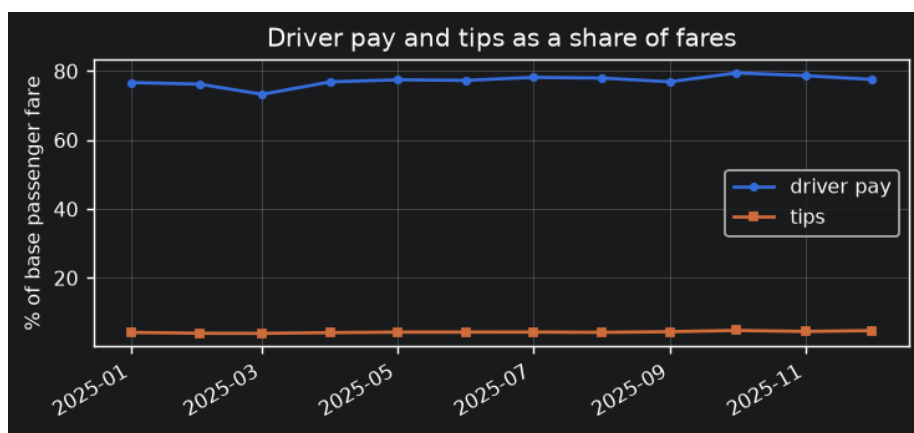


Figure A7. Driver pay and tips as a share of base passenger fare, by month.

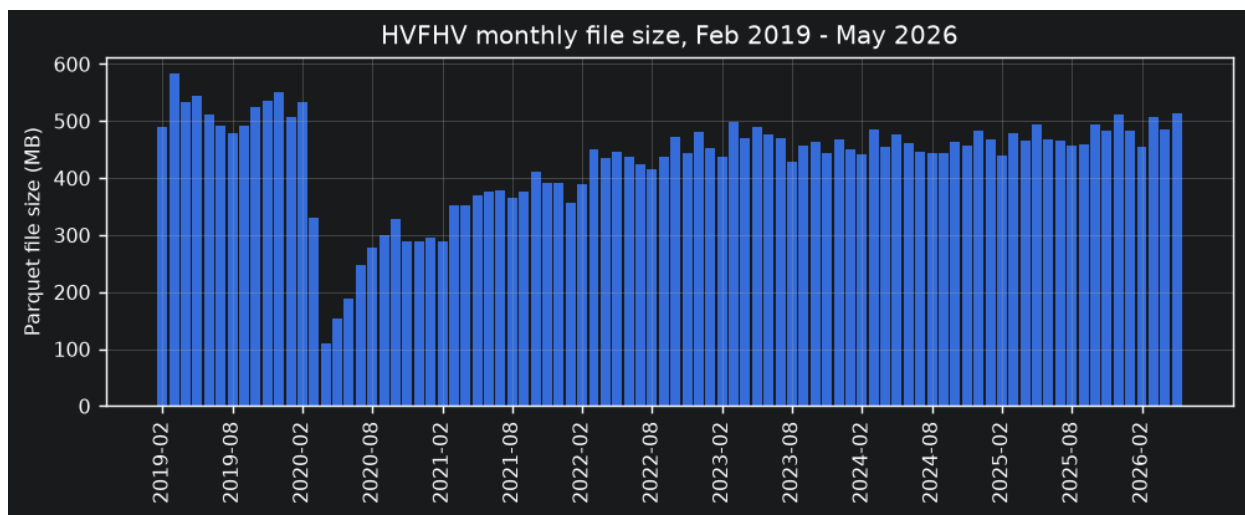


Figure A8. Monthly file size across the archive. The 2020 collapse is stark, but file size is a poor proxy for trip count: 2025 files are comparable in size to 2019 files while carrying one additional column, and direct row counts place 2025 volume at 94.6% of 2019.